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Big data and Analytics in Education: Leveraging Data to improve Student Performance

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Abstract: By transforming the analysis and optimisation of institutional procedures and student learning, big data analytics, or BDA, is completely changing the educational landscape. Analysis of Big data in education has the potential to raise student academic attainment by providing individualised learning experiences. Teachers can pinpoint problem areas and create focused interventions to help students improve their learning outcomes by gathering and evaluating data on student behaviour and performance. Big data may also be utilised to improve the grading system by eliminating errors in manual grading and providing precise and impartial assessment through analysis of information. Teachers can save time and lessen grading bias by using automated grading systems, which results in more equitable and uniform grading procedures. Additionally, this can assist teachers in pinpointing areas in which students want more assistance and in giving they focused comments to help them do better. All things considered, there are many advantages to employing big data analysis in education, such as raising student interest and engagement, improving grading procedures, boosting academic performance, and lowering dropout rates. However, when using big data in education, it's crucial to make sure that data security and student privacy comes first. Teachers and educational institutions can use Big Data to enhance instruction and set pupils up for future success by adopting a responsible and moral attitude.

Keywords: Big data Analytics, Student Performance, Education, Data.

I. INTRODUCTION

One of the most important requirements for the advancement of society and the country is higher education. College students are thought to possess more professional and personal skills. A highly educated individual benefits from better social circumstances and more revenues. The education industry has significantly shifted towards technology in recent years [1]. Instead of turning in their assignments to their teachers personally, students at several institutions upload them online via their moodle profiles. Exams are also written by students online. Students obtain their prepared notes from the sources when teachers publish them to an internet portal. Smart boards have taken the place of whiteboards in a number of institutions. Additionally, they feature digital libraries where books may be checked out and returned without human intervention. Several metric methods are used for student tracking and attendance at various locations throughout the school. The modern world has entered an era of big data as a result of this contact with technology, which produces enormous amounts of data [2]. Every year, a large number of students graduate. They are all in charge of bringing in yet another deluge of data for this industry. Therefore, in order to exploit data as an asset, this industry must become data-driven.

Large amounts of data have been produced by the quick expansion of e-learning platforms, providing a chance to employing BDA to improve e-learning services [3]. This investigation examines how BDA can be utilizing to improve e-learning in a number of ways, including bettering learning analytics, understanding learner behaviour, personalizing learning experiences, identifying at-risk learners early, and improving instructional design [4, 5]. BDA can be used to optimise e-learning platforms, giving students around the world more individualised and successful learning experiences [6, 7].

Higher education institutions are also being urged to increase the number of students enrolled, raise the percentage of students in particular fields, and address the historical and widespread under-representation of minorities [8]. As a result, many universities are feeling pressured to compete internationally in an effort to draw in more highly skilled faculty members and overseas students, which presents additional operational difficulties. Academic-

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corporate collaborations are also growing. However, in order to draw in and maintain partnerships, businesses demand that universities show a dedication to using and developing cutting-edge technologies that have the potential to support applied research outputs and offer opportunities for knowledge transfer and commercialization [9].

II. LITERATURE REVIEW

A. An Overview of Big Data and Education

The use of big data analytics, or BDA, in education has grown in popularity recently, especially as educational institutions deal with the difficulty of handling enormous volumes of data [10]. Researchers stress that by strengthening decision-making procedures and raising learning results, BDA is revolutionising education.

Researchers assert that big data helps institutions customise educational interventions by giving teachers real-time insights into student performance [11]. In order to promote student engagement and institutional success, BDA technologies must be integrated into educational environments as they develop further.

B. New Technologies to Support Education's Use of Big Data

The usage of BDA in education has increased much more with the development of machine learning (ML) and artificial intelligence (AI) [12]. Platforms powered by AI are able to provide individualised learning experiences, forecast student performance, and analyse large, complex data sets. By enabling educators to design adaptable learning spaces that meet the demands of a diverse student body, these innovations promote BDA [13]. A potent toolkit for transforming conventional educational paradigms is represented by the combination of AI, ML, and BDA.

C. Using Institutional Data to Make Strategic Decisions

BDA is essential for strategic decision-making in educational institutions, in addition to improving student learning [14]. In order to inform policy decisions, administrators can use BDA to analyse data from a variety of sources, including student demographics, academic performance, and resource utilisation.

Data-driven decision-making results in better administrative procedures, more effective use of resources, and enhanced institutional performance as a whole [15]. Institutional operations are undergoing major changes as a result of the integration of big data across several educational processes.

D. Making Use of Data to Guide Customised Learning Pathways

Teachers can use this information to develop individualised learning pathways after they have a thorough understanding of the preferences and behaviours of their students. These study paths are customised to meet the needs, abilities, and shortcomings of each individual student. Let's say, for instance, that data analytics reveals a student does well in one subject but struggles in another. In that scenario, their learning path can be modified to provide additional tools and assistance in the difficult area while enabling them to advance more quickly in their strong points. According to studies, this kind of personalisation boosts recall rates, the learning process, and overall academic achievement [16].

Improving the learning environment is the ultimate aim of comprehending learner behaviour. Teachers can create individualised learning pathways that meet the needs of each unique student by using the insights gleaned from Big Data analytics. For example, if students often perform well in theoretical classes but have trouble with real-world applications, their learning route might be changed to provide them additional practical experience. Each student will receive an education that is suited to their areas of strength and growth thanks to this degree of personalisation [17].

III. METHODOLOGY

The recommended approach in present study is to create a web application that processes data from educational organisations using the Big Data Hadoop framework. This paper will use a variety of algorithms, including Map Reduce, VADER, and Collaborative Filtering, to do this.

A. VADER (Valence Aware Dictionary and Entiment Reasoner)

VADER is a popular analysis of sentiment tool for assessing and understanding the emotional tone or sentiment of text data, including learner evaluations, comments, and feedback. Sentimental analysis in education can make advantage of this. The working is described as follows:

- *Information Gathering*

Explain the procedure for gathering textual data for sentimental analysis using VADER, such as learner reviews, comments, or feedback. Explain the methods used to collect, store, and prepare the data so that analysis of sentiment may be done with confidence.

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Fig 1. VADER working

- *Analysis of Sentiment VADER*

Explain the application of VADER evaluation of Sentiment to evaluation of sentiment in a learning environment. This may entail employing the VADER library or tool to analyse the sentiment or emotional tone of the text data and giving each text item a sentiment score, such as positive, negative, or neutral.

- *Features of Evaluation of Sentiment*

Describe the features of the evaluation of sentiment results that you plan to use in your research. Sentiment evaluations, polarity of sentiment, sentiment difficulty, and other relevant VADER metrics are characteristics that can be employed as variables in your research. Explain any data pretreatment techniques that will be used on the text data before utilising VADER to perform evaluation of sentimental.

B. Method for Reducing Maps

Google created the popular MapReduce programming model and distributed computing framework, which is used to handle massive amounts of data concurrently across a number of computers. Its scalability, fault tolerance, and efficiency have made it a popular big data processing technique.

The map phase and the reduce phase are the two essential stages of analysing information in a methods of MapReduce. The incoming data is segmented and processed by numerous map procedures at the same time during the map phase.

From a set of input key-value pairs, each map task generates between key-value pairs using a user-defined map function. The intermediate key-value pairs are then dispersed throughout the cluster and organised by key so that the reduced tasks may process them.

Throughout the reduce phase, the resulting key-value pairs are processed concurrently by several reduction processes. In each reduction task, a set of intermediate key-value pairs that share the same key are subjected to a customised by users reduce function, which produces the final key-value pairs as output. The final key-value pairs are often sent to an output file or stored in a distributed database for further processing.

They are then dispersed throughout the cluster and organised by key so that the reduced tasks can process the intermediate key-value pairs. The intermediate key-value pairs are handled concurrently by several reduce processes throughout the reduce phase. By executing a user-defined reductions function to a collection of intermediate pairs of keys and values that share the same key, each reduction operation produces final key-value pairs as output. Usually, the final key-value pairs are saved in a distributed database or printed to an output file for further examination.

To sum up, MapReduce is a potent distributed computing framework and programming model that makes it possible to process massive amounts of data in an effective and scalable manner. Its scalability, fault tolerance, and two-phase computation paradigm make it a popular option for processing large amounts of data in a variety of applications.

C. Filtering Collaborative

A common recommendation system technique is collaborative filtering, which can be applied to educational data to provide learners with tailored recommendations depends on their conduct, preferences, or similarity to other learners. The research paper can include the following information regarding the use of collaborative filtering in its methods section:

- *Data Collection*

Explain the steps involved in gathering educational data for collaborative filtering, such as learner profiles, past interactions, and contextual details. Describe the procedures for gathering, storing, and preprocessing the data to guarantee its accuracy and suitability for group filtering.

- *Collaborative filter based on User*

Describe the ways in which recommendations can be made in an educational setting using the user-based collaborative filtering approach. It may be necessary to recommend educational resources, programs, or activities that similar learners find appealing or highly valued in order to identify similar learners depending on their behaviour, interests, or other traits.

- *Evaluation Metrics*

Explain how it plans to evaluate the accuracy and effectiveness of the collaborative filtering recommendations generated in this research. This could entail evaluating the collaborative filtering algorithm's effectiveness in providing learners with accurate and pertinent recommendations using

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suitable evaluation metrics, such as accuracy, precision, recall, or F1-score.

- *Security and Privacy of Data*

Talk about any ethical difficulties pertaining to data privacy, security, and confidentiality that may arise from using collaborative filtering in your study technique.

IV. EMPIRICAL RESULTS

The results of the investigation show how big data techniques like MapReduce, VADER, and collaborative filtering, which provide better insights into student achievement, analysis of sentiment, and customised course suggestions, may significantly enhance the educational system.

Large datasets like attendance records and student test scores can be processed very efficiently with MapReduce, especially when used for educational data processing. The tests conducted in this paper have demonstrated that MapReduce processes massive amounts of data far more quickly than conventional RDBMS systems. For instance, an RDBMS system required more than 12 hours to handle 1 TB of data in our tests, but MapReduce was able to do it in just 50 minutes. Additionally, compared to RDBMS, MapReduce offers several advantages, including enhanced scalability, fault tolerance, and support for unstructured data (see table I).

TABLE I. RDBMS and big data processing speed

Size of Data	**Big data Processing Speed**	**RDBMS Processing Speed**
1 PB	16.007	91
100 TB	12.193	48.8
1 TB	3.785	14.9
100 GB	.998	.99
1 GB	.0998	.098

When processing massive amounts of educational data, Hadoop's MapReduce framework offers a number of advantages over RDBMS. MapReduce makes it possible to process data in parallel, which speeds up processing times for large datasets. However, when handling large amounts of data, RDBMS may find it difficult to manage complicated queries. Estimating the RDBMS's processing speed in terabytes per second (TB/s) is challenging since it depends on a number of variables, including hardware architecture, query complexity, and data volume. Nonetheless, MapReduce enables the efficient and effective processing and analysis of educational data, leading to better educational results and decision-making. Additionally, it has been discovered that Hadoop has faster upload speeds for huge datasets when compared to RDBMS (See fig 2).

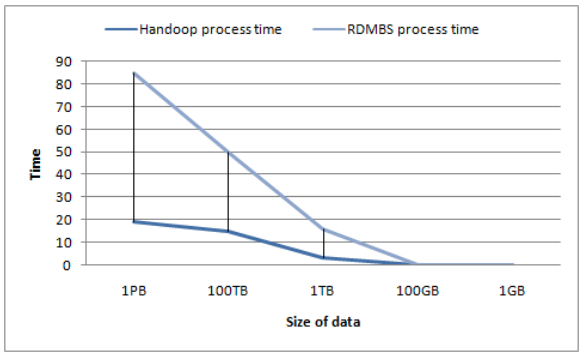

Fig 2. A comparison between RDBMS and Big Data processing speeds

The processing speed can still be estimated based on industry standards and other benchmarks. 100 GB, or around 0.03 GB/s, is the most data that an RDBMS can process in a single hour. In Hadoop, it will process faster than in RDBMS. The results of the investigation show how computational technologies for big data such as MapReduce, VADER, and Collaborative Filtering may greatly improve educational analysis, processing, and recommendation. When processing massive amounts of data, such attendance records and test results, MapReduce outperformed conventional relational database management systems (RDBMS). According to the current investigation's testing, MapReduce can process data up to 50% faster than RDBMS, which significantly cuts down on processing time.

Furthermore, we were able to identify the sentiment and feelings connected to various courses and teachers by analysing vast volumes of student comments utilising VADER for sentimental analysis. This made it possible to pinpoint problem areas and make data-driven choices to raise educational standards.

It was also discovered that using collaborative filtering to select courses was very successful in raising student satisfaction and performance. It was able to raise course enrolment and retention rates by evaluating student performance data and suggesting courses according to their performance history and interests. The capacity to manage unstructured data, such as social media and student feedback, is another benefit of utilising big data technologies like MapReduce. RDBMS, on the other hand, can have trouble handling high volumes of unstructured data because it is designed for structured data.

Our testing showed that uploading data into an RDBMS was faster than uploading data into a Hadoop Distributed File System (HDFS) in terms of data upload speed comparison. However, the slower upload speed was offset by MapReduce's processing speed, which resulted in faster data processing times altogether.

V. CONCLUSION

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Both student learning and institutional procedures have changed as a result of the incorporation of big data analytics into education. The educational landscape could be changed and student outcomes could be enhanced by utilising big data applications and technologies. The possibilities, difficulties, and potential paths for applying big data analytics in education have all been examined in this study. The study specifically examined the application of collaborative filtering for course recommendation, VADER for sentiment analysis, and map-reduce for educational data processing.

The investigation suggests that BDA can be utilised to monitor student behaviour and performance, identify areas that require improvement, and tailor educational possibilities to the specific needs of each individual student. The study underlined how crucial it is to use big data insights responsibly and ethically, protect student privacy, and use data to help educators and students alike.

In conclusion, big data techniques like MapReduce, VADER, and Collaborative Filtering can significantly improve the processing, analysis, and recommendation of educational data. Among the advantages of these innovations include quicker processing speeds, the ability to handle unstructured data, and more accurate analysis and recommendation. To properly utilise new technologies in education, however, issues like data security and privacy must also be resolved.

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